

MERN Developer Practical Assessment (4–5 Hours)

Objective

- Build a Kanban-style task management system.
- Focus on logical data handling, state management, and API design.

Core Functional Requirements (Mandatory)

- JWT-based authentication.
- Boards → Ordered Lists → Ordered Cards.
- Drag & drop cards across lists with order persistence.
- Card update (title required, due date, status).
- Search/filter cards (status or keyword).
- Users access only their own boards.

Frontend – Required Pages & UI Scope

- 1. Auth Pages: Login & Register (form validation required).
- 2. Dashboard Page: List all boards + Create/Delete board.
- 3. Board Detail Page:
 - - Display lists horizontally.
 - - Each list shows ordered cards vertically.
 - - Add/Edit/Delete list.
 - - Add/Edit/Delete card (modal or inline form).
 - - Drag & drop cards within/across lists.
 - - Search/filter UI on top.
- UI must handle loading states, errors, and empty states.
- Protected routes for authenticated users only.

Logical Implementation Focus

- Maintain consistent ordering when moving cards.
- Update only affected items during reorder (avoid full reindex).

- Prevent invalid operations (wrong list/board).
- Cascade delete: board → lists → cards.
- Clean and consistent API response format.
- Handle edge cases (invalid token, bad input, invalid date).

Backend Expectations

- Structured architecture (routes/controllers/models/middleware).
- Input validation and centralized error handling.
- Ownership validation on all operations.
- Pagination for card listing.
- Proper HTTP status codes.

Optional Enhancements

- WebSocket real-time updates (else polling).
- Board member invitation.
- Activity log for card movements.
- Unit tests for reorder logic.

Submission

- Working frontend and backend.
- README with setup steps.
- Environment variables configured properly.